Automations

🚨 1. AUTOMATION: ALERT NEARBY USERS

🧠 Core Idea

When emergency is triggered:

👉 Find users within radius (e.g., 2 km)

👉 Send them instant notification

⚙️ HOW IT WORKS (END-TO-END)

```
1. App sends emergency → Backend
2. Backend fetches user location
3. Backend finds nearby users
4. Send push notifications to them
```

📍 STEP 1: STORE USER LOCATION

Every user app should periodically send location:

```
POST /update-location

{
  user_id,
  latitude,
  longitude
}
```

👉 Store in DB (MongoDB recommended)

📍 STEP 2: FIND NEARBY USERS (GEO QUERY)

Use **MongoDB GeoSpatial Query**

Example logic:

```
Find users where distance < 2km
```

MongoDB supports:

- `2dsphere index`
- `$near` query

📡 STEP 3: SEND ALERT (FCM – Firebase)

Use:

👉 **Firebase Cloud Messaging (FCM)**

Flow:

```
Backend → FCM → Nearby Users Devices
```

Notification Content:

```
{
  title: "🚨 Emergency Nearby",
  body: "Someone needs help near you",
  data: {
    location,
    emergency_id
  }
}
```

📱 STEP 4: USER RECEIVES ALERT

App shows:

- Map with victim location
- Button: “Help / Navigate”

📞 2. AUTOMATION: CALLING + MESSAGING

Now second part — this is **very powerful for demo**

📲 OPTION A: SEND SMS (EASIEST)

✅ Use: Backend API (BEST)

Use:

👉 **Twilio API**

Flow:

```
Backend → Twilio → SMS to contacts
```

Message Example:

```
"Emergency! I need help. My location: https://maps.google.com/?q=lat,long"
```

📞 OPTION B: AUTO CALL (VERY IMPORTANT LIMIT)

❌ Reality:

You CANNOT auto-call silently from backend

✅ What you CAN do:

Option 1: Trigger call from app

```
Backend → Send command → App → Initiates call
```

Use:

- Android Intent (`ACTION_CALL`)

Option 2: Use Twilio Voice Call (BEST FOR DEMO)

Backend makes automated call:

- Plays message:

> "This person is in danger, location shared via SMS"

Flow:

```
Backend → Twilio Voice API → Call contacts
```

🔁 3. COMPLETE AUTOMATION FLOW (COMBINED)

🚨 Emergency Triggered

```
1. App → Backend (emergency data)

2. Backend:
   → Find nearby users
   → Send FCM notifications

3. Backend:
   → Send SMS via Twilio
   → Trigger voice call
```

```
4. App:
   → Start live tracking
```

🧱 4. TECH STACK FOR THIS PART

Backend:

- Spring Boot (since you know Java)

Database:

- MongoDB (for geo queries)

Notifications:

- Firebase Cloud Messaging (FCM)

SMS/Call:

- Twilio API

🔥 5. HACKATHON SHORTCUT (IMPORTANT)

If time is less:

DO THIS MINIMUM:

- Nearby users → FCM ✅
- SMS → Simple API or even mock ✅
- Call → Simulate button press in app ✅

🎯 WHAT YOU SAY TO JUDGES

👉

“We built an automated emergency pipeline where once a distress signal is validated, our backend instantly identifies nearby users using geospatial queries and triggers multi-channel communication including push notifications, SMS, and voice calls without any manual intervention.”

⚡ PRO TIPS (THIS WILL IMPRESS JUDGES)

✅ Show real-time demo:

- One phone triggers emergency
- Another phone receives alert

✅ Show map:

- Nearby user location

✅ Show SMS:

- With live location link